



AER1416: Numerical Methods for Uncertainty Quantification

University of Toronto Institute for Aerospace Studies
<http://utias.utoronto.ca/~pbn>

Assignment 1 (35pts)

Due March 5, 2026

1. (14pts) Consider the quadratic function in **P1**.
 - (a) (8pts) Derive closed form expressions for the mean and variance of f . In your derivation, consider the general setting with d variables. Show all key working steps and provide the final expressions.
 - (b) (4pts) Apply the Monte Carlo method to estimate the mean, variance and cumulative distribution function of f when $d = 10$ and $\gamma = 0.5$. Perform convergence studies and compare the mean and variance obtained using Monte Carlo to the analytical solution.
 - (c) (2pt) Using the batch Monte Carlo approach, demonstrate the central limit theorem at work for this particular problem. Consider the case when it is sought to approximate the variance when $d = 10$ and $\gamma = 0.5$.
2. (5pts) Approximate the mean and variance of the displacement of the system in **P2** as a function of time over the time-horizon $[0,10]$ seconds using the Monte Carlo method.
3. (5pts) Implement a Monte Carlo method for approximating the mean and variance of the components of the vector $\mathbf{u}$ defined in **P3**.
4. (6pts) Devise an approach to approximate the mean and variance using a first-order approximation, i.e., by approximating the function of interest as a linear function of the random variables and subsequently using the results we derived in class for linear functions of random variables. Evaluate the accuracy of the linear approximation against the analytical expressions for **P1** and Monte Carlo results for **P2** and **P3**. [1pt for **P1**, 2pts for **P2**, and 3pts for **P3**].
5. (5pts) Apply the control variate approach to **P1** with $d = 10$ to estimate the mean of f . Choose the linear control variate $a_0 + \sum_{i=1}^d a_i x_i$. Evaluate the performance of this variance reduction approach in comparison to the standard Monte Carlo method for two cases (i) $\gamma = 0.5$ and (ii) $\gamma = 0.1$. Discuss your results and explain the reasons for the observed performance.

Here are some guidelines for your calculations:

- For each test problem, conduct numerical studies to establish the sample size required to obtain accurate approximations.
- Calculate the standard error of the approximations for each test case and report all results with 99% confidence intervals.

Submission guidelines

- Your report must not exceed 12 pages (with at least 10pt font type and single spacing). Dedicate one section to each problem and present results in tables and/or figures as appropriate. Discuss the results obtained and briefly outline your conclusions. While answering 1(a), show all key working steps and clearly define all variables used in your derivation.
- Submit an **electronic copy** of your report in PDF format, **documented** MATLAB/Python scripts and **input files** (if any). You should include a file named “README” outlining how the scripts should be run. Upload a single **tar** or **zip** file containing all files to Quercus. You are expected to verify the integrity of your tar/zip file before uploading it to Quercus.
- All submissions must comply with the AER1416 Collaboration Policy posted on Quercus.